Social Network

[ASSIGNMENT 02]

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Course: [Social Network]
Submission Date: [23-Dec-2025]

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1 Theoretical Questions

1.1 Question 1: Structural Index

1.1.1 Part (a) Explanation of each function

1. $f_1(G)$ (*Network efficiency*): Average reciprocal distance between all node pairs. **High value** = nodes are closer on average (small-world property). **Low value** = larger distances or disconnected pairs.

2. $f_2(G)$ (*Weighted local closure*): Measures triangular reinforcement of attention flows. **High value** = strong triadic closure in interaction patterns. **Low value** = weaker local closure.

3. $f_3(G)$ (*Degree distribution uniformity*): Negative normalized entropy of degree distribution. **High value** (closer to 0) = degrees more uniform. **Low value** (more negative) = degrees more unequal.

4. $f_4(G)$ (*Average online hours*): Average hours worked online per employee. **High value** = longer online hours. **Low value** = shorter online hours.

5. $f_5(G)$ (*Average weighted degree*): Average connection strength per node. **High value** = stronger connections. **Low value** = weaker connections.

1.1.2 Part(b) Structural Index determination

A **Structural Index** depends only on network connectivity (topology/weights), not on external attributes.

- **Structural Indices**: f_1, f_3, f_5
  - f_1: based on shortest paths (topological)
  - f_3: based on degree distribution (topological)
  - f_5: based on connection weights (structural in weighted graphs)
- **Not Structural Indices**: f_2, f_4
  - f_2: depends on attention proportions p_{ij} (behavioral allocation)
  - f_4: depends on hours worked $H(v)$ (external attribute) [3]

Node	Closeness Centrality
1	0.467
2	0.467
3	0.467
4	0.400
5	0.452
6	0.452
7	0.467
8	0.424
9	0.452
10	0.438
11	0.412
12	0.400
13	0.519
14	0.368
15	0.275

Table 1: Closeness centrality values for nodes.

1.2 Question 2: Distances and Neighbors

1.2.1 Part (a): Network of 15 nodes

Scenario 1-Radio broadcast station (minimize maximum distance): Find the graph center (smallest eccentricity). Node 13 has eccentricity 3 (farthest nodes: 4, 8, 12 at distance 3).
No node has eccentricity ¡3, so 13 is a center and has highest closeness (0.519). Best node: 13.
Scenario 2-Two stores (sequential location): First store A should be placed centrally to capture many nodes; node 13 is best.
Given A at 13, store B should be placed to maximize its own capture.
Checking ties: With A=13, B=3 captures nodes 1,3,4,7 — more than other choices after calculation. Best: A at 13, B at 3.
Scenario 3-Bookstore (minimize total distance): The graph median has the highest closeness centrality: node 13 (closeness 0.519). Best node: 13.
Note: The closeness centrality values reported in Table 1 were computed in Python (tools.py) using the NetworkX library, a widely used toolkit for graph analysis.

1.2.2 Part (b) Ring network with central hub C

Let total nodes n = hub C + m ring nodes, $m = n - 1$.
1. Closeness centrality for node C: All shortest paths from C to others = 1. Sum of distances = m.
Closeness $C_C = \frac{n-1}{m} = \frac{m}{m} = 1$.

(Assuming $C(i) = \frac{n-1}{\sum_j d(i,j)}$).

2. Closeness centrality for a ring node: Let ring node v:

- To hub C: distance 1
- To 2 ring neighbors: distance 1 each
- To other $m-3$ ring nodes: distance 2 (via hub)

Sum of distances $= 1+1+1+2(m-3) = 2m-3 = 2n-5$.

Closeness $C_{\text{ring}} = \frac{n-1}{2n-5}$.

3. Betweenness centrality for node C:

Only shortest paths between ring nodes at ring distance ≥ 3 pass uniquely through C; ring distance 2 pairs split equally. Unnormalized betweenness:

$$B_C = \frac{(n-1)(n-5)}{2}, \quad n \geq 6.$$

(c) Tree with edge separating regions of sizes n_1, n_2

Let $S_1 = \sum_j d(1,j)$, $S_2 = \sum_j d(2,j)$.

Edge (1,2) splits tree into region 1 (size n_1 including node 1) and region 2 (size n_2 including node 2), $n = n_1 + n_2$.

For v in region 1, $d(2,v) = d(1,v) + 1$.

For v in region 2, $d(1,v) = d(2,v) + 1$.

Let $D_{11} = \sum_{v \in \text{region1}} d(1,v)$, $D_{22} = \sum_{v \in \text{region2}} d(2,v)$.

Then $\sum_{v \in \text{region1}} d(2,v) = D_{11} + n_1$, and $\sum_{v \in \text{region2}} d(1,v) = D_{22} + n_2$.

So: $S_1 = D_{11} + (D_{22} + n_2)$

$S_2 = (D_{11} + n_1) + D_{22}$

Subtract: $S_2 - S_1 = n_1 - n_2$.

Closeness: $C_1 = \frac{n-1}{S_1}$, $C_2 = \frac{n-1}{S_2}$.

Then:

$$\frac{1}{C_1} - \frac{1}{C_2} = \frac{S_1 - S_2}{n-1} = \frac{-(n_1 - n_2)}{n-1}.$$

Thus:

$$\frac{1}{C_1} + \frac{n_1}{n-1} = \frac{1}{C_2} + \frac{n_2}{n-1}.$$

The given formula in the question has denominator n instead of $n-1$; that would require closeness defined as $C_i = \frac{n}{\sum d(i,j)}$ — likely a typo.

1.3 Question 3: Stress Centrality

Node stress centrality $C_s(v)$ only counts shortest paths where v is an intermediate node ($s \neq v$, $t \neq v$). Edge stress centrality $C_s(e)$ counts all shortest paths through edge e, including those where v is an endpoint.

For in-edges to v, $C_s(e_{\text{in}})$ includes paths ending at v ($t = v$), which are excluded from $C_s(v)$. For out-edges from v, $C_s(e_{\text{out}})$ includes paths starting at v ($s = v$), also excluded from $C_s(v)$. Therefore:

$$C_s(v) \; < \; \sum_{e \in \Gamma(v)} C_s(e)$$

because the edge sum counts extra paths where v is an endpoint, and double-counts paths through v (once in the in-edge, once in the out-edge).
The exact relationship is that $C_s(v)$ equals the sum over pairs $(e_{\text{in}}, e_{\text{out}})$ of the number of shortest paths using both edges consecutively via v, excluding cases with $s = v$ or $t = v$.

2 Implementation Questions

2.1 Question 1: Structural Analysis of Political Power

2.1.1 Part (a): Power Geometry

1. Centrality Calculations
In this part, we calculate three centrality measures: Degree Centrality, Eigenvector Centrality, and Closeness Centrality. These measures help in understanding the structural position of the nodes in the network.
Degree Centrality: This is the number of direct connections a node has. Nodes with a higher degree are considered more influential in terms of direct connectivity. Degree centrality was normalized between 0 and 1 for comparison across nodes.
Eigenvector Centrality: This centrality measure takes into account not only the number of connections a node has, but also the quality (importance) of those connections. It assigns higher scores to nodes that are connected to other highly-connected nodes.
Closeness Centrality: This measure evaluates how close a node is to all other nodes in the network. Nodes with high closeness can reach all other nodes with fewer steps, making them more "efficient."
Here are the Top 10 nodes for each centrality metric:

Degree Centrality	**Eigenvector Centrality**	**Closeness Centrality**
Manfred Weber	Katarina Barley	Barack Obama
Joachim Herrmann	Arno Klare MdB	Michael Roth
Katarina Barley	Katja Mast	Niels Annen
Arno Klare MdB	Christian Petry	Tanja Fajon
Katja Mast	Heike Baehrens	Malcolm Turnbull
Barack Obama	Klaus Mindrup	Mariano Rajoy Brey
Angela Merkel	Michelle Müntefering	Achim Post
Niels Annen	Niels Annen	Peter Tauber
Martin Schulz	Johannes Schraps	Dietmar Nietan
Sir Peter Bottomley MP	Sigmar Gabriel	Hillary Clinton

Table 2: Top 10 Nodes by Centrality Measures

2. Gap Analysis

We performed a gap analysis between Degree Centrality and Eigenvector Centrality by plotting the normalized degree against the normalized eigenvector centrality. The plot highlights the correlation between these two centrality measures and identifies the outliers that show low degree but high eigenvector centrality.

The following nodes were identified as outliers (with low degree but high eigenvector centrality):

Tanja Fajon

Mariano Rajoy Brey

Hillary Clinton

This scatter plot,figure 1, highlights nodes that exhibit low degree but high eigenvector centrality, marked in red. These nodes are significant because although they have relatively few direct connections (low degree), they are highly connected to other influential nodes in the network, indicating that their influence comes not from direct connectivity but from being connected to powerful nodes. This can be seen as a form of indirect influence.

In social networks, these nodes can act as bridges or conduits, facilitating communication or access to other powerful nodes, even if they don't have a large number of direct connections. This behavior is particularly relevant in political networks, where influence may stem from connections to key individuals rather than sheer network size.

These outliers suggest the importance of considering not just the number of connections a node has but also the quality of these connections. The red-marked nodes could be used strategically to influence network dynamics, despite their lower degree.

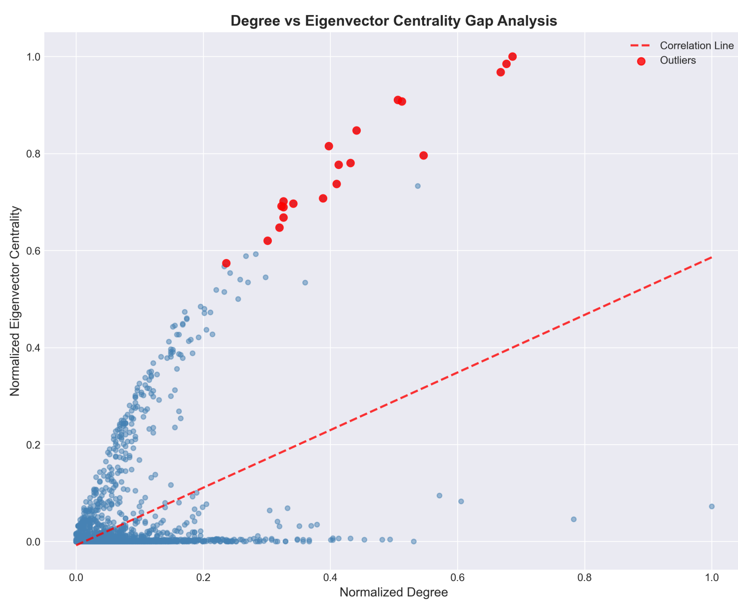

Figure 1: Degree vs Eigenvector Centrality Gap Analysis. Outliers identified as low-degree, high-eigenvector nodes are marked in red.

3. Three-Way Case Study

We selected three politicians exhibiting Low Degree (ranked outside the top 100) but High Eigenvector (ranked within the top 50):

* Andrea Nahles
* Christian Flisek
* Olaf Scholz

These individuals were analyzed for their Closeness Rank, indicating their dependence on neighbors:
High Closeness: The node is at the geometric heart of the network, with more independence.
Low Closeness: The node is marginally attached to the core, dependent on powerful neighbors.
The selected politicians and their closeness status are as follows:

Politician	Degree Rank	Eigenvector Rank	Closeness Status
Andrea Nahles	102	30	Low (Marginal Attachment)
Christian Flisek	110	45	Low (Marginal Attachment)
Olaf Scholz	113	36	Low (Marginal Attachment)

Table 3: Centrality Ranking and Status of Selected Politicians

2.1.2 Part (b): Information Bottlenecks

1. Betweenness Centrality

We calculated Betweenness Centrality, which measures how often a node acts as a bridge along the shortest path between two other nodes. The top 10 nodes with the highest betweenness centrality are as follows:

Politician	Betweenness Centrality	Degree Rank
Barack Obama	0.4095	6
Hillary Clinton	0.0633	141
Justin Trudeau	0.0626	27
Manfred Weber	0.0623	1
Angela Merkel	0.0602	7
Malcolm Turnbull	0.0579	35
Peter Tauber	0.0485	38
Betinho Gomes	0.0476	120
Niels Annen	0.0467	8
Boris Johnson	0.0382	665

Table 4: Comparison of Betweenness Centrality and Degree Rank among Politicians

2. Rank Gap Analysis: Bridges vs Hubs

We analyzed the structural difference between Bridges (high betweenness, low degree) and Hubs (high degree, low betweenness). The Bridges identified in the top 10 betweenness centrality are listed in 4

Hillary Clinton
Justin Trudeau
Malcolm Turnbull
Peter Tauber
Betinho Gomes
Boris Johnson
These nodes serve as key connectors between communities, while Hubs have many connections but remain within their own communities.

2.1.3 Part (c): Power in Local Structures

1. Closeness Centrality
The top 10 nodes with the highest Closeness Centrality are as table 5.

Politician	**Closeness Centrality**
Barack Obama	1.0000
Michael Roth	0.8258
Niels Annen	0.8180
Tanja Fajon	0.7962
Malcolm Turnbull	0.7864
Mariano Rajoy Brey	0.7850
Achim Post	0.7838
Peter Tauber	0.7749
Dietmar Nietan	0.7631
Hillary Clinton	0.7628

Table 5: Top Politicians by Closeness Centrality

2. Degree vs Closeness Scatter Plot
A scatter plot was generated to explore the relationship between Degree and Closeness Centrality. The nodes in the Top-Left Quadrant (high closeness but low degree) were identified as efficient monitors. These nodes include:
* Tanja Fajon
* Mariano Rajoy Brey
* Hillary Clinton
Figure 2, visualizes the relationship between degree and closeness centrality. The nodes in the top-left quadrant, which have high closeness but low degree, are identified as "efficient monitors". This indicates that these nodes are well-positioned to access any part of the network quickly, despite having relatively few direct connections.
Nodes with high closeness centrality can reach any other node with fewer steps, meaning they are crucial for the rapid spread of information across the network. Politicians in this category might not control many direct connections, but they are positioned in key places that allow them to gather and disseminate information efficiently.
These nodes could serve as critical players in the communication or decision-making process,

especially in cases where efficient monitoring or rapid responses are needed. Analyzing their role could help in understanding how influence operates in a more subtle, less direct manner within a network.

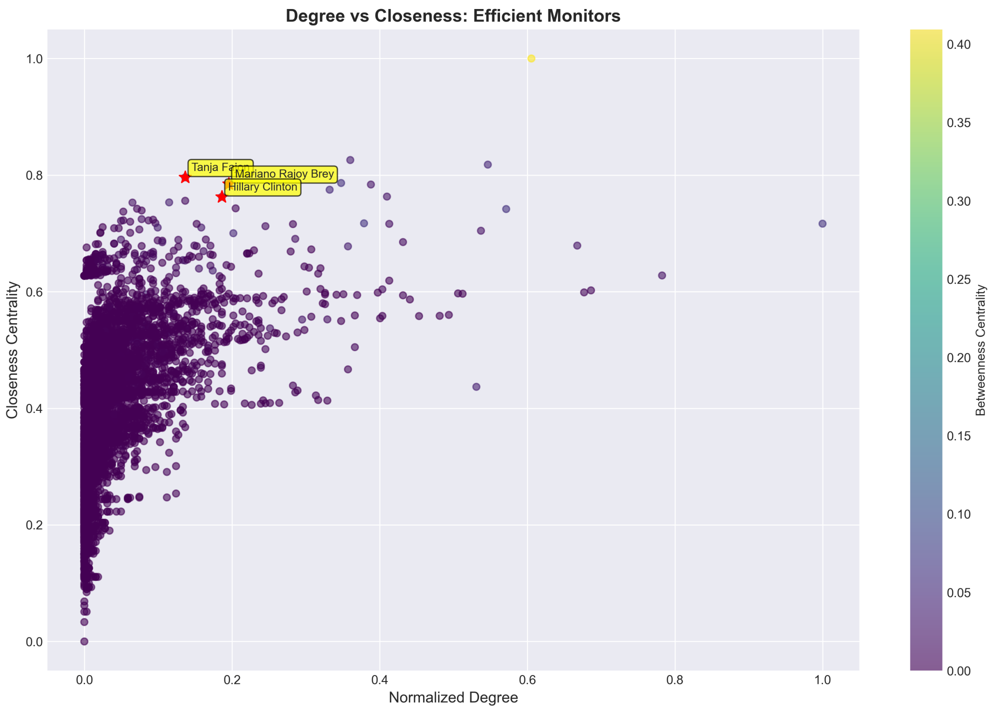

Figure 2: Degree vs Closeness: Efficient Monitors. The top-left quadrant highlights politicians with high closeness and low degree.

3. Ego Network Visualization

The ego network of Tanja Fajon was visualized, showing how this node connects to others in the network. The network is bridge-like, as her neighbors are dispersed rather than forming a dense cluster. The visualization is shown in Figure 3 below:

2.1.4 Part (d): Bonacich Power Dynamics

1. Bonacich Power Regimes

The Bonacich power is calculated under three regimes: Neutral ($\beta 0$), Supportive ($\beta 0$), and Suppressive ($\beta 0$). The largest eigenvalue was found to be 60.6431, with a corresponding $\beta_m ax$ value of 0.0165.

2. Slope Chart (Rank Trajectories)

Figure 4 shows the ego network of Tanja Fajon, who has high closeness and low degree. The ego network visualization demonstrates how Tanja's neighbors are scattered throughout the network rather than clustered together, indicating a "bridge-like" structure.

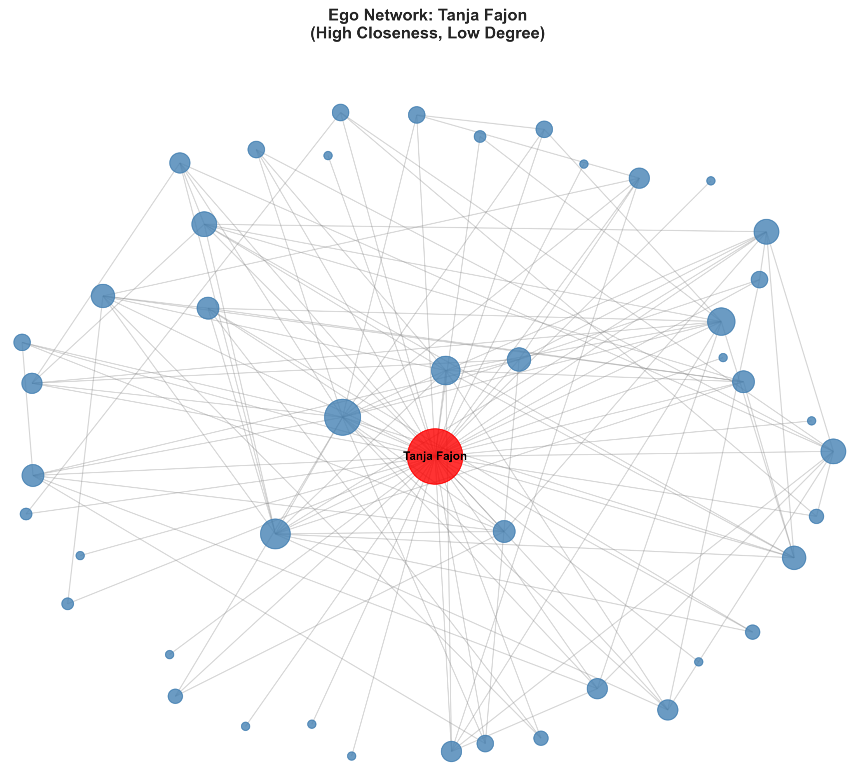

Figure 3: Ego Network: Tanja Fajon (High Closeness, Low Degree). The network shows a bridge-like structure with dispersed neighbors.

A node with high closeness but low degree can serve as a key connector in a network, providing access to multiple other nodes without necessarily being heavily connected to them. This scattered structure is indicative of a node that has access to different parts of the network without being embedded in any specific community. In political networks, such individuals often hold the power of "information flow" rather than direct control.
Tanja Fajon's position in the network might not give her control over many connections, but her ability to quickly access any part of the network could give her substantial political leverage. This highlights the power of connectivity in terms of access rather than quantity.
The slope chart, figure 4, shows how nodes' ranks shift under different Bonacich power regimes (Neutral, Supportive, and Suppressive). This figure demonstrates how the ranking of nodes changes based on the value of β (the influence factor).
Bonacich power is a measure of a node's relative power, considering both its own degree and the degrees of its neighbors. The shifts in rank across different β values indicate how the power dynamics in the network are affected by the surrounding nodes. When β is high (supportive regime), nodes that are connected to influential others rise in rank, whereas in a suppressive regime (low β), the influence of central nodes is minimized.
This analysis reveals the dynamic nature of political power and influence. The Bonacich power model helps us understand how a node's power might change depending on whether it is in a supportive or suppressive environment. It also underscores the importance of not only

looking at individual node measures but considering how nodes interact with their neighbors.

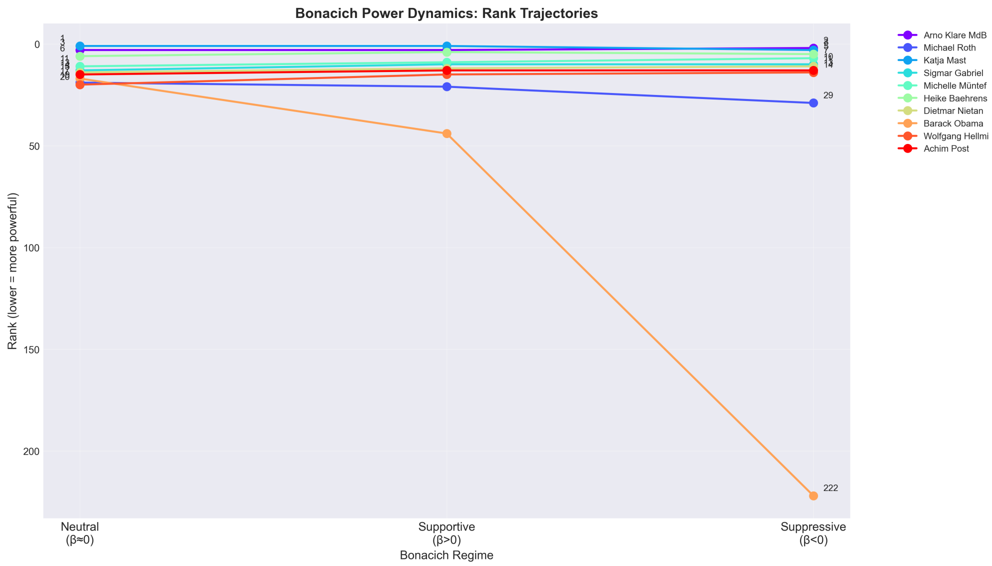

Figure 4: Bonacich Power Dynamics: Rank Trajectories.

3. Role Classification Based on Rank Shifts

Nodes were classified into three categories based on the changes in their Bonacich power ranks:

Power Amplifiers: Nodes that rise significantly in rank with $\beta 0$. Power Inhibitors: Nodes that drop in rank with $\beta 0$. Stable Actors: Nodes with minimal changes in rank.

Code and tools Explanation

The code utilizes several powerful Python libraries for network analysis and visualization. "pandas" is used for reading and processing the CSV files containing node and edge data, as well as manipulating the data to compute centrality metrics and store them in a DataFrame for easier analysis. "networkx" is the primary library for constructing the graph and calculating various centrality measures like degree, eigenvector, closeness, and betweenness, which are essential for analyzing the influence of nodes in the network. "numpy" supports matrix operations, particularly for Bonacich power centrality, where the inverse of the adjacency matrix is computed. "scipy" is employed to compute eigenvalues of the adjacency matrix via the 'eigsh' function, which is used in the calculation of Bonacich power. For visualization, "matplotlib" is used to generate plots such as scatter plots for degree vs eigenvector centrality, slope charts to analyze rank changes, and ego network visualizations to understand local node structures. The tools were chosen because of their efficiency and compatibility in handling graph-based data, calculating centrality measures, and producing insightful visualizations.

2.2 Question 2: Comparative Analysis of Ranking Algorithms in Directed Networks

2.2.1 Part (a): Ranking Comparison (HITS vs. PageRank)

1. Calculation and Mapping

We begin by applying both HITS and PageRank algorithms to the `Wiki-Vote` dataset, which represents the voting network for Wikipedia administrators.

HITS Algorithm

The HITS algorithm computes two types of scores for each node: *Authority Scores* and *Hub Scores*. The authority score reflects the importance of a node based on the number and quality of its incoming links, while the hub score reflects the importance of a node based on its outgoing links.

PageRank Algorithm

The PageRank algorithm computes the rank of nodes based on the network's link structure. It uses a damping factor ($\alpha = 0.85$) to model the behavior of a random surfer navigating the network. The PageRank score is updated iteratively using a power iteration method until convergence.

Rank Mapping

After computing the authority and PageRank scores, we rank the nodes by their respective scores. The node with the highest score is assigned Rank 1. These ranks are used to create a log-log scatter plot to compare the rankings of HITS and PageRank [1].

This scatter plot in Figure 5 compares the authority ranks from the HITS algorithm with the PageRank ranks. Ideally, a perfect correlation would show a straight line along the diagonal (y = x). The plot highlights how some nodes deviate significantly from the diagonal, indicating that the two ranking algorithms differ in their assessment of node importance.

The HITS algorithm distinguishes between two types of nodes: hubs (which have many outgoing links) and authorities (which have many incoming links). In contrast, PageRank only considers the global link structure, not differentiating between hubs and authorities. The nodes deviating from the diagonal highlight how these two algorithms evaluate nodes differently: nodes that have high authority but low out-degree (HITS) will rank differently from PageRank, which values global influence over local connections.

This divergence analysis is crucial for understanding how different ranking algorithms can highlight different aspects of network influence. The outliers in the plot show nodes that may have a substantial influence in one algorithm but not in the other, emphasizing the need to choose the appropriate algorithm depending on the specific context of network analysis.

2. Divergence Analysis

Nodes that deviate significantly from the diagonal line $y = x$ in the scatter plot are identified. These are the nodes where the rankings from HITS and PageRank differ significantly. The structural reasons behind these divergences are analyzed by examining the local and global patterns of incoming links.

The divergence is mostly due to local endorsement (HITS) versus global influence (PageRank). For example, Node 3762, which has no incoming links, has a higher rank in PageRank

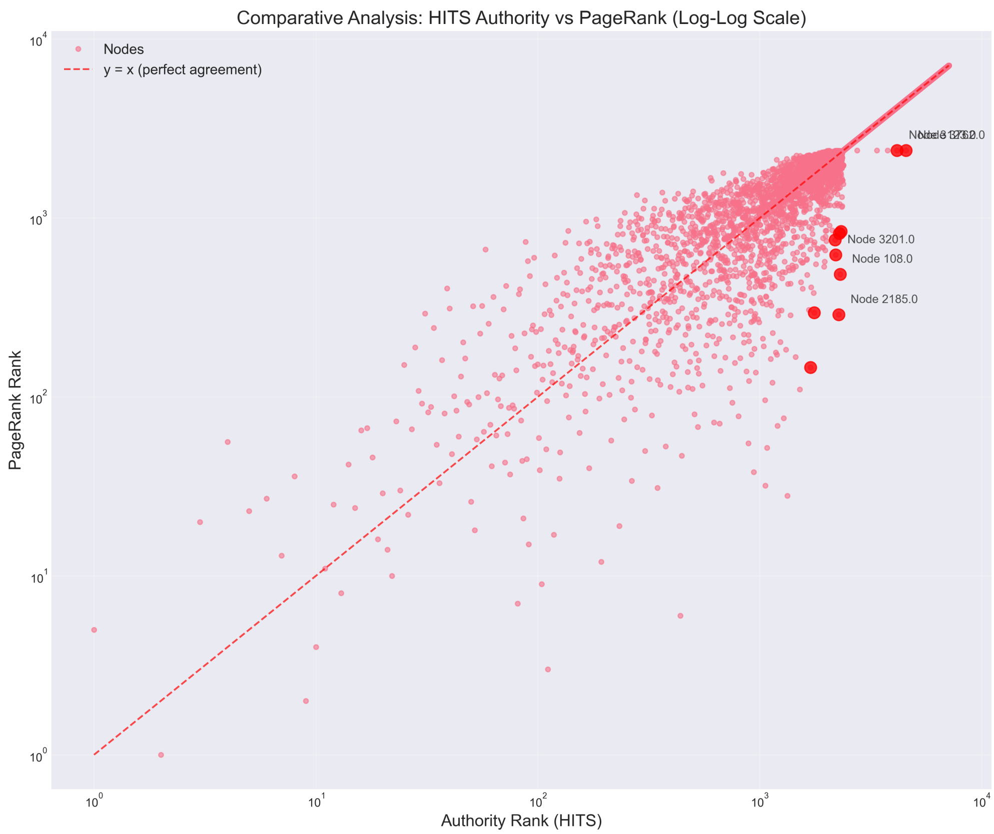

Figure 5: Comparative Analysis: HITS Authority vs PageRank (Log-Log Scale)

Node	**Authority Rank**	**PageRank Rank**	**Rank Difference**	**In-degree**	**Out-degree**
3762	4566	2386	2180	0	8
2185	2271	288	1983	10	0
108	2303	485	1818	11	2
3123	4151	2385	1766	0	3
3201	2199	622	1577	4	0

Table 6: Divergent Nodes Analysis

due to endorsements from influential nodes, while its HITS rank is lower because it lacks local authority. Nodes such as 2185 and 108 are ranked higher in PageRank because they are endorsed by other influential nodes in the network, although their own incoming links (authority) are not substantial.

2.2.2 Part (b): Rank Stability Analysis

1. Simulation (PageRank Sensitivity)
The PageRank algorithm is executed across a range of damping factors α from 0.50 to 0.99. The rank trajectories of the top nodes and those identified as outliers are tracked. A line chart is generated to visualize how the ranks change as α increases.

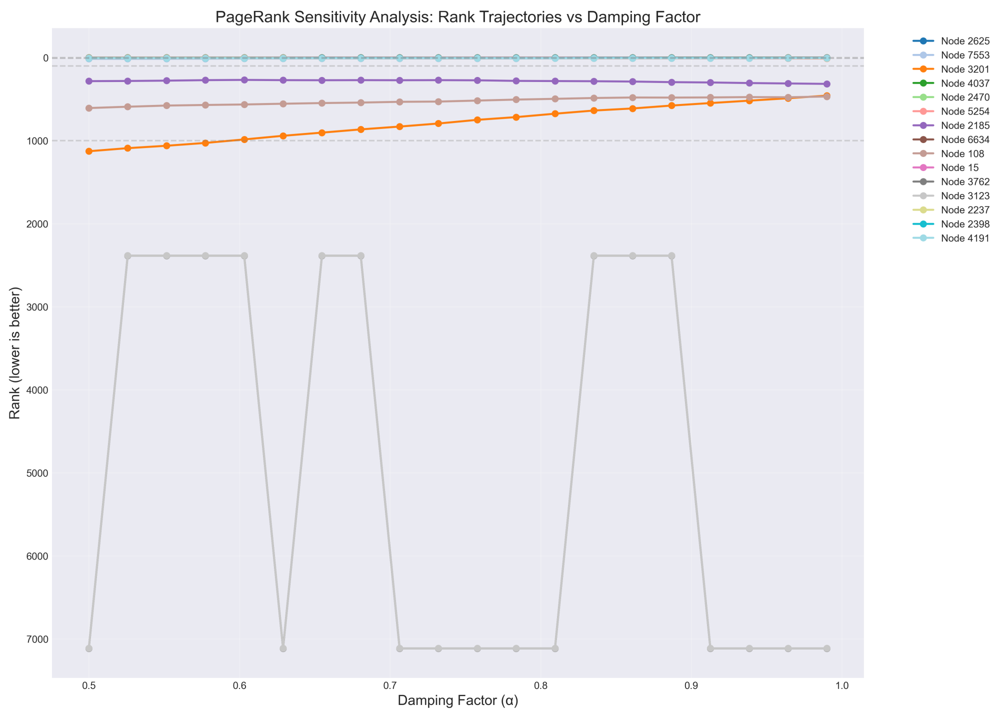

Figure 6: PageRank Sensitivity Analysis: Rank Trajectories vs Damping Factor

This line chart , figure 6, shows how the ranks of various nodes change with different values of the damping factor (). The damping factor in PageRank controls the likelihood of random surfing in the network, affecting how much influence each link has on the node's rank.
Some nodes, like Node 2625, show stable ranks across varying values, indicating that their importance in the network is relatively consistent. Other nodes, such as Node 2470, experience significant fluctuations in rank, suggesting that their influence is more dependent on random surfing than on the network's link structure.
This sensitivity analysis provides insight into how different nodes in the network respond to changes in the PageRank algorithm's damping factor. Nodes with stable ranks are more robust in terms of their influence, while nodes with fluctuating ranks are more sensitive to

changes in the network structure or the random teleportation factor. This can help identify which nodes have consistent influence regardless of algorithmic changes.

2. Trajectory Interpretation

The rank trajectories of various nodes show how their ranks change with different values of α. We classify nodes based on the following trends:

- **Stable**: The node's rank remains relatively constant regardless of the damping factor.
- **Improving**: The node's rank improves (i.e., decreases) as α increases.
- **Declining**: The node's rank worsens (i.e., increases) as α increases.

For example, Node 2625 shows a stable rank, indicating its importance is consistent across various damping factors. In contrast, Node 2470 experiences a declining rank, suggesting that its influence is more dependent on random teleportation rather than the link structure.

Rank Correlation Metrics The Spearman rank correlation and Kendall's Tau were computed to measure the agreement between HITS and PageRank rankings:

$$\text{Spearman Rank Correlation} = 0.9930, \quad \text{Kendall's Tau} = 0.9571$$

These high correlation values suggest that, despite the differences in algorithmic approach, both HITS and PageRank largely agree on the relative importance of nodes in the network.

Top 10 Nodes Overlap

Out of the top 10 nodes ranked by HITS and PageRank, 4 nodes overlap (40.0%). This indicates that while both algorithms identify some of the same influential nodes, they also assign high ranks to different nodes based on their respective methodologies.

Rank Divergence Statistics

The following statistics were computed for the rank divergence between HITS and PageRank:

$$\text{Average Rank Difference} = 107.70, \quad \text{Maximum Rank Difference} = 2180.00$$

This large maximum rank difference suggests that while the algorithms generally agree, there are nodes for which the rankings differ significantly. For instance, Node 3762 has a rank difference of 2180, indicating a substantial disagreement between HITS and PageRank for this node [4].

These comparisons highlight the nodes that are ranked highly by one algorithm but not the other, providing insight into how the algorithms evaluate node importance differently.

Code and Tools Explanation

The code leverages several powerful tools to analyze and compare the HITS and PageRank algorithms applied to the Wiki-Vote dataset. NumPy is used for efficient numerical computations, particularly matrix operations essential for both algorithms. Pandas organizes the node rankings into DataFrames, enabling easy manipulation and comparison of results. Matplotlib and Seaborn are employed for visualizing the comparative rankings and the sensitivity

Node	HITS Rank	PageRank Rank
3456	14	42
737	15	24
2565	8	36
2535	20	29
4712	18	46

Table 7: Nodes in HITS Top 20 but Not in PageRank Top 20

Node	HITS Rank	PageRank Rank
7553	104	9
1186	193	12
6946	233	19
7620	91	15
5254	22	10

Table 8: Nodes in PageRank Top 20 but Not in HITS Top 20

analysis of PageRank with varying damping factors, enhancing the clarity and aesthetics of the plots. NetworkX constructs and manipulates the directed graph, computing important graph properties like in-degrees and out-degrees, which are crucial for the ranking analysis. SciPy handles sparse matrix operations, optimizing memory usage for large datasets. Finally, Warnings are suppressed to maintain clean output during intensive computations. Together, these tools facilitate the loading, computation, visualization, and analysis of node rankings, offering insights into the behavior of these two ranking algorithms in directed networks [2].

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